

Reflections



Image 1: A photograph taken on the final day of fieldwork at my study site. Never much of a fan of selfies, but working alone in the field made occasional self-documentation feel necessary, both as proof of presence and as a memory of a place and time I know I will dearly miss in the years to come.

Coming to Lesbos was serendipitous, as I mentioned; it was never part of the grand plan, and yet here I was, doing terrestrial ecology on this most enchanting of islands. In many ways, it was different from Gran Canaria for me, principally because I was in the midst of company that felt much more wholesome and endearing than what I had experienced before, and that alone changes one's experiences and overall stay. This island hosts not only the usual Erasmus+ students, many of whom seem to arrive with the sole intention of amassing ever more blackout stories to later recount with gusto and pride, but also volunteers working at the refugee camp, people whose perspectives on life often felt profoundly different and whose experiences across the myriad organisations operating there made them an absolute pleasure to talk to and get to know. When I stepped into my fourth semester, I was already slightly late in starting my thesis, since I had decided to take yet another ecology course at RUG called *Island Biology*—and if one is familiar with my programme, this is not a course I could afford to miss. Thus, I was already under a mild time crunch when most of my peers had begun their thesis fieldwork. I felt as if a ticking time bomb was constantly hovering nearby, though in hindsight that pressure worked in my favour, forcing me to plan properly. The project, initially proposed as something grounded in chronobiology and sociality, gradually morphed

into something entirely different. However, the literature I had read before arriving on Lesbos was not wasted; instead, it became a springboard for thinking more broadly about activity rhythms in field conditions. Through discussions with chronobiologists at RUG, I came to realise that what I was proposing was certainly possible in principle, but within the constraints of a thesis timeframe it would have been infeasible and unnecessarily tedious. It would have required collecting bees across different levels of sociality for comparison, which is particularly tricky in a region where I had perhaps overestimated the existing natural history work. If I remember correctly, it was during the first online meeting with both Victor and Thomas that I realised the topic was already taking a different shape from what I had initially proposed. I was slightly taken aback at first, but given enough time to reflect before arriving on Lesbos, I realised that behavioural thermoregulation was not a bad direction either. I was keen to work in bee-behavioural ecology anyway, as it is the field I would like to pursue in my future doctoral studies. Gaining fieldwork experience in this area felt ideal, and it made more sense to install cameras in the field rather than work in a controlled lab environment, as I had originally proposed. My first in-person meeting with Thomas took place on 27 February, shortly after I arrived on the island and was still being drawn into its bee-guiling charm. I would admit that for the first two solid weeks, I had very little clear idea of what I would actually be doing here.

But gradually, through reading literature and digital ideation over emails with Thomas and Victor, I finally zoned in on two parts of my thesis: one behavioural ecology component, which would form the bulk of the work, and a second supplementary part involving physiological measurements of thermal limits to support trends observed in the first. It is funny to look back and see how much this kept changing and refining over time, because during my very first trip to the Malemi Hotel with Moritz, we were very happy to have found what I sincerely believed were the nests we were looking for, having never seen a ground-nesting bee nest before—though I am now fairly certain they were abandoned ant nests. I was at such a naive phase, almost an amateur with bees, that even honeybees looked bee-zarre to me, let alone other bee species that I could scarcely bee-gin to identify. I stared through my binoculars as they carried pollen in their little pollen baskets, and I was bee-wildered to realise that I had never observed a bee like this before, and that my binoculars were equally good at observing insects as birds; it was all far more overwhelming than I had expected, because insects are literally everywhere. When I was later told by Thomas that the actual study site was outside the hotel premises, I still stuck to my instincts and believed they might have missed these nests in previous years, especially since it seemed so convenient to me that they were within the hotel grounds. I thought it would make everything easier,



Image 2: (A) Overview of the multi-camera setup used during behavioural sampling across both field sites. Cameras included Thanasis' action camera for focal observations at Site 1 and my iPad for wide-angle coverage at the same site. At Site 2, Sveta's DSLR was used for focal observations and my GoPro for wide-angle coverage. Michael's handheld camcorder was used for opportunistic recordings of the field sites and surrounding behaviours. An additional camera angle was initially recorded using my phone but later replaced with Michael's GoPro due to the need to preserve the phone for other field purposes; this dataset was ultimately not used due to time constraints and redundancy with the primary GoPro recordings, particularly given the highly multi-angled distribution of nest entrances at Site 2. Michael's second GoPro was used to record the chill coma recovery assays. In total, 7–8 devices were in rotation, making battery management, charging logistics, and SD card limitations (often only 8 GB, requiring frequent swapping to prevent overwriting) a daily but demanding challenge that kept me constantly prepared for things going wrong. (B) Camera setup used at Field Site 1 during behavioural ecology sampling. Thanasis' action camera (visible in white) was positioned for focal recordings of selected nest entrances, while my iPad was used for wide-angle behavioural observations across the wider nesting aggregation. Note the discarded wiring and metal fencing strewn around the setup.



Image 3: (A) Chill coma recovery assay being conducted by me and Isabella on a honeybee individual. Isabella recorded body temperatures and recovery durations while I held the thermal camera. The icebox used for cold exposure is visible on the left, alongside the insect net and cotton balls used to seal plastic vials after bee collection. (B) Close-up of surface temperature measurements being taken from a bee positioned ventral side upwards during righting duration and body temperature recordings. (C) Experimental setup used for the chill coma recovery assay. All equipment was placed inside a thermally insulated chamber (a polystyrene box) used to minimise external temperature fluctuations during trials. The setup includes two stopwatches: one used to record the duration individuals spent in the icebox, with checks initially conducted minute-by-minute and later at 30-second intervals once movements subsided (icebox not shown). A thermocouple was used to measure ambient temperature, while a thermal camera (not switched on in this configuration) was intended for surface temperature measurements. A second stopwatch was used to record trial durations. Flight recovery was scored only when individuals were able to initiate true flight outside the chamber, excluding brief or incomplete “mock flights,” to maintain a consistent metric across both bee species.



Image 4: One of the first potential nests I initially sent to Victor, which I believed to be *H. scabiosae*. In retrospect, the nest entrance appears excessively surrounded by twigs and debris and was most likely an unused ant nest located within the hotel premises.

as I would not have to lug a 30 kg battery through tall vegetation every day, and since the cameras also needed to be transported daily because I could not leave them outside due to the risk of rain, it all felt more manageable. When I stayed at the hotel in March (25–30) in Skala Kallonis, eagerly waiting to begin preliminary data collection, Thomas and his PhD student Antigone helped me set up the cameras they had used in previous seasons. Although the setup seemed extremely cumbersome on the surface, requiring daily transport through dense vegetation without an assistant despite my attempts to recruit several juniors and Erasmus students, the real issue emerged when we inspected the site Thomas had in mind and found that the soil was waterlogged and there were no nests in sight. We were all disappointed, as it looked completely different compared to the dry summer conditions when the landscape was barren, and for me it was especially disheartening, since I had been hoping to begin sampling during this week of fieldwork. I decided to rig all six cameras at night while speaking to my sister to calm my nerves as panic built up, and I was genuinely proud when I managed to get everything working, although that moment quickly turned against me the next morning when I attempted to record what I think were “ant” nests, still believing they were bee nests, and discovered that the small monitor used to check camera angles and focus was not functioning. At that point, frustration, dis-bee-lief, worry, and panic all set in at once; I was anxious because not only did I not have any confirmed nests to record, as I had never observed bees entering the sites I had selected, but I also had no working camera system even if I

did. I video-called Thomas and Antigone, and Thomas decided to order a replacement system, similar to the existing rear-view car camera monitor, but it would only arrive on Monday, while I was originally due to leave on Tuesday. I therefore decided to use the remaining time for natural history scouting around the site, and that was when luck finally struck during a wandering survey of the unmanaged pasture behind the swimming pool fence of the Malemi Hotel, where I encountered the first large nesting aggregations. At the time, I did not even have an insect net, so using only a plastic vial, and hoping I would not get stung, as I initially believed these sweat bees stung as well, I carefully collected individuals for identification using AI before sending them to Victor and Thomas. This entry is the first in my field journal and is quite reflective of what my first proper field day looked like:

26th March 2026; 8:44 am Inside hotel premises: No visible activity Outside hotel premises: Wet soil, vegetation, promising

10:40 am Figuring out how this monitor works. I have set up the cameras, but haven't started recording yet.

7:05 pm

Horrible morning. I'm not the most technologically challenged person; at times I'm surprisingly rather adept with such things. But this morning I came close to admitting defeat, bee-reft of words to express my astonishment. I still do not have the faintest idea why some equipment that worked perfectly yesterday refused to work today. The only explanation I can fathom is that I used the same battery plugs as input for the "small screen," which is really an antique reverse camera monitor for a car from the 1980s, and that the battery could not handle the multiple connected systems running from it simultaneously.

Oh well. That at least forced me to explore the grounds a bit more, and lo and behold—I found *H. scabiosae* :)

Beautiful, shiny, and flying as though on acid—I initially thought it was a rabid wasp. ChatGPT told me it was a hoverfly. Gemini reacted with ardour, maintaining that it is *H. scabiosae*. ChatGPT failed. Having found enough nests for four cameras (originally there were six, but two had broken wires that could not be soldered back on), I was satisfied after all. My thesis did not sink; the ship is still afloat, though desperately in need of repair. The water is out for now.

After activity dwindled into oblivion around 3 pm, I ordered a heavy meal at Skala Kallonis and watched Balkan terrapins while my phone battery nearly died. I went back to check on the bees around 6 pm and was rudely driven out by pestilent hordes

of swarming mosquitoes. Now seated on this rather large bed with clean sheets, the air conditioner relegated to a heater, and a dinner buffet awaiting me, I feel as close as one can get to a privileged researcher from a first-world country conducting “extremely tedious and gruelling” fieldwork in whatever remains of the wilderness in a Mediterranean second-world country.

The only stark difference is that I still do not know whether I am paying for this out of my own pocket or not. I really hope the personal travel grant responds affirmatively tomorrow.

Worried and a bit lonesome,
Amey



Image 5: The unmanaged pasture meadow surrounding the nesting sites during the early phase of fieldwork. Initially, the nesting areas were dominated by Persian clovers, but over the following weeks the landscape changed rapidly, gradually becoming bee-sieged by yellow and white flowering patches due to blooming dandelions, chamomiles, and *Raphanus raphanistrum* (wild radish). Toward the later stages of sampling, scarlet pimpernels also began appearing near the nesting sites, making the meadow feel as though it transformed subtly every few days throughout the season.

And it did work out in the end—I managed to secure the personal travel grant after writing a proposal for this thesis:

Dear Members of the Review Committee,

I am writing to apply for a personal travel grant to support the expenses associated with the fieldwork for my master’s thesis. I am a second-year student in the Islands and Sustainability Erasmus Mundus master’s programme at the Faculty of Spatial Sciences, University of Groningen. I am currently based on the Greek island of Lesbos, where I am conducting my master’s thesis research on bee chronobiology and behavioural



Image 6: The very first nesting aggregation I encountered at the field site. After pushing through a dense thicket of grass and vegetation, this sudden barren patch of exposed soil appeared almost unexpectedly, and it was the moment I finally realised what the nesting aggregations actually looked like in the field. The soil seemed perfect: sandy, bare without vegetation, and dry, with countless *L. malachurum* nests scattered across the patch, which immediately gave me hope that the field season might actually work out after all.



Image 7: The first site where I observed multiple *H. scabiosae* nests clustered together, alongside numerous individuals actively foraging and collecting pollen and nectar from Persian clovers.



Image 8: Initial Site 2, located approximately five metres from Site 1, appeared to function as an extension of the main nesting aggregation and contained several additional isolated nests. At the time, I speculated that these may have represented separate nesting aggregations with different foundresses, and I sent these observations to Victor on 26 March. This site later became important during fieldwork, as I used Michael's camcorder here to record opportunistic behaviours and collect B-roll footage for a mini-documentary summarising my thesis fieldwork during the 10-minute observation periods conducted throughout the main sampling phase between 15–24 April.



Image 9: My final Field Site 2, which consisted of a mound-like “village” of nesting aggregations dominated by *H. scabiosae* and *L. malachurum*, alongside several other bees including species from the genera *Chelostoma* and *Andrena*. During the evenings, the site gradually became shaded by the adjacent wall, and I initially suspected that this evening shadow may have influenced activity patterns and possibly even the timing of nest-closing behaviour.



Image 10: This was one of the first moments when I realised that the bees had only recently begun excavating and establishing their nests, as indicated by freshly dug earth and the surrounding turret of loose sand (a tumulus). Activity peaked around midday, with the nesting aggregation surrounded by ants, bee flies, beetles, pill bugs, and numerous other arthropods that I could scarcely identify, possibly including brood parasites and kleptoparasites. I also observed, for the first time, female bees cautiously protruding either their heads or abdomens from nest entrances whenever another insect or nestmate approached too closely. By around 15:00, when I sent this image to Victor on 26 March, activity had already declined substantially.



Image 11: The red line indicates the route Victor used to access the back of the hotel and the approximate area where nesting aggregations had been located during previous summers when he conducted observations there. The purple markings represent the locations I initially believed to be the study sites I had found. In reality, these were entirely incorrect; the actual nesting aggregations described in the manuscript were situated much closer to the swimming pool and adjacent pathways. For a considerable time, however, this mistaken mental map persisted in my head, which in hindsight feels like a fitting reminder of just how disoriented and visually overwhelmed one can be at the very beginning of fieldwork. Sent to Victor on 27 March.

thermoregulation in affiliation with the University of the Aegean. At present, I am living in the city of Mytilene, which is approximately 43 km from my field site near Lake Kalloni, in the area adjacent to the Malemi Hotel. Previous studies in this location have been conducted by Dr. Victor Gonzalez (University of Kansas) and my supervisor, Dr. Thomas Tscheulin (University of the Aegean). Their work identified nesting aggregations of *Halictus scabiosae*, a species belonging to the Halictidae bee family, in the vicinity of the Malemi Hotel.

Halictus scabiosae is an obligately eusocial bee species that often reuses the same nests each year to form colonies typically consisting of 6–120 individuals. In some populations (for example in Switzerland), colonies show distinct castes and hierarchy, with one breeding female and several worker-forager females. As the season progresses, the filial generation (the queen's offspring) replaces the original workers after they are driven out of the nest. This process transforms the colony from an initially semi-social structure with multiple breeding females into a primitively eusocial colony characterised by overlapping generations, cooperative brood care, and reproductive division of labour. Displaced females may drift and establish nearby, resulting in nesting aggregations that make this species particularly suitable for behavioural monitoring.

For my thesis, I plan to deploy six cameras near these nests for two consecutive days each week. After each sampling period, I will return to Mytilene to review the footage and

ensure that the recordings were successful before the next week of data collection. I intend to begin fieldwork around 25th April, when female bees typically begin emerging and becoming active, and continue until the end of June. My thesis defence, which will take place in Gran Canaria (my mobility island), is scheduled for 30th June. Because of this schedule, I will need to travel to Kalloni each week for two days of fieldwork. The field site is relatively remote and the nearest available accommodation is the Malemi Hotel, which is located very close to the nesting area, effectively within the hotel's grounds. Staying there allows easy access to the site and the staff are also familiar with our research and help us store equipment safely. The cost of accommodation is €30 per night. Over a two-month period, this amounts to approximately €60 per week, €240 per month, and €480 in total. In addition, I estimate about €150 for food during the two months and approximately €80 for bus transportation between Mytilene and Kalloni. Altogether, a grant of around €650 would allow me to cover these essential fieldwork expenses.

I have planned to spread my sampling across multiple weeks rather than conducting it over a short continuous period. This approach helps reduce the influence of short-term weather conditions that might otherwise confound behavioural observations. Collecting data across several weeks will also allow me to capture subtle seasonal transitions from spring to early summer, which may influence bee behaviour. The camera recordings will later be analysed by scoring behavioural activity in 10–30 minute intervals (the exact methodology is still being refined). In addition to recording nest-entry and nest-exit frequencies, I will estimate the number of individuals active around each nest and approximate foraging bout durations where possible. I will also record environmental variables such as temperature, humidity, and light intensity (to determine whether nests are shaded or exposed). Another component of the analysis will involve documenting the presence of parasites or predators, including bee flies, cuckoo wasps, gold wasps, ichneumon wasps, *Sphecodes* bees, and velvet ants. A key behaviour of interest in my study is nest-closing behaviour, which often peaks after midday when parasite activity increases and bee foraging activity declines. We aim to investigate whether this behaviour functions primarily as an anti-parasite strategy or as a form of behavioural thermoregulation. Summer temperatures in Lesvos can reach around 40 °C, so temporarily sealing the nest may help regulate internal nest conditions. Observing this behaviour earlier in the season, when parasite pressure is still relatively low, may help clarify whether thermoregulation plays an important role, or whether the behaviour is primarily anticipatory and defensive.

Due to the limited timeframe before my thesis defence, my research focuses on a single

well-documented site. Collecting sufficient replicates from one location will allow me to obtain robust data within the available time. Ideally, I would also survey additional sites with similar environmental characteristics (sandy or loamy soils, thermophilic vegetation, and nearby floral resources), but this is not feasible within the current timeframe. Previous studies have also focused on single nesting aggregations, which provides a useful basis for comparison. Within the site itself, I will still be able to examine subtle environmental variation, for example by comparing nests in shaded versus exposed locations. In addition to nest-closing behaviour, I may also document other behaviours such as nest guarding, in which the queen positions herself at the nest entrance to prevent parasites or predators from entering. Dr. Victor Gonzalez has already collected data on activity patterns during the summer months, and my research will complement these observations by focusing on earlier seasonal activity and particularly on nest-closing behaviour before parasite activity peaks.

Although I receive an Erasmus Mundus stipend, a personal travel grant would greatly help me cover the additional costs required for this fieldwork. This support would allow me to carry out consistent weekly observations and collect the data necessary to complete my thesis successfully. If awarded the grant, I would be happy to share documentation of the fieldwork and acknowledge this funding support in my thesis and related outputs.

Thank you very much for considering my application. I really appreciate this opportunity and the time you have taken to review my application. I eagerly look forward to your response!

Sincerely,

Amey Danole

I think this was the first instance of me writing an actual proposal, not a mock one, for a real topic I would be working on. It is interesting to look back at the timelines mentioned here: quite ambitious, and the expenses I had estimated are still broadly close to the actual ones, although some, such as transportation, were way off, and the comparative analysis part of the story had not even been conceived yet. It is a shame to look back and realise that I could not afford to spend much more time at the site to obtain longitudinal data as I had originally proposed, since budget constraints, and later, during data analysis I realised quite harshly, time constraints, would not allow me to collect data over too many days. Additionally, the original plan to record parasites and predators near the site was not very successful because, although I had good camera quality and resolution in the end, there was simply so much happening that quantifying everything within the limited analysis timeframe seemed nearly impossible. But I am

rushing ahead; let us go back to my March stay, when the only humans I spoke to were at night, when a German tourist group hunting orchids around the island gathered for dinner. The rest of the time, I spoke with the countless African and South American parrots, rabbits, cockerels, squirrels, and South American coatis that the hotel owner kept on the premises. Through my days of observation, this week became instrumental to the questions I would later develop during the remainder of my thesis, and I am almost glad the cameras did not work and that I did not end up studying ants in despair and desperation instead of bees, which had been my plan B.

At 8:59 am on 27 March, I observed a bee-fly hovering over a *H. scabiosae* nest, witnessed through binoculars as it repeatedly attempted to flick its larvae while a single guarding individual tried to repel it, or at least that is how I interpreted it at the time, with some help from Gemini. It was also the first time I noticed *L. malachurum* lurking around the nesting site (confirmed by Victor on 27 March), and this was when I realised for the first time how different the nesting sites looked in the morning compared to the evening. My field journal entry at 9:23 am on 27 March reads: “The nests stand out like bullet holes. Each species marring the earth every which way, unique and architecturally incomprehensible.” At 11:18 am, after the briefest of rain showers, I wrote: “But a lot of the ‘bullet holes’ are smoothed out now and it is really difficult to tell if there are nests here in the first place.” I had a lot of free time between observation bouts, and when I needed a break from constantly looking through the binoculars and writing, especially poetically, it helped pass the time. I immediately noticed how important light intensity was, as bees would come out of their nests during brief bursts of sunlight when clouds moved aside, for example at 9:37 am: “whenever there is a moment of sunny respite, I see some activity in the bees.” By 11:18 am, I was already considering a comparative idea after observing several small bees becoming active shortly after brief rainfall, which made me wonder whether these species might have different thermal tolerance limits, with some potentially emerging under lower light conditions, whereas *H. scabiosae* might require a threshold temperature before becoming active. I decided to record videos on my iPad and phone simply to practice analysis, which turned out to be a good decision because this was when I realised how good those recordings were. Being horrible at identifying plants (apologies, Theophrastus) and suffering from chronic plant blindness, I attempted a basic plant survey of the site and recorded insect taxa as well, which quickly became a herculean task due to the sheer diversity within such a small patch of land. I was also very naive at the time, under the belief that bees carrying pollen in the morning might somehow be different species, as their intensely yellow appearance seemed to bee-lie what they actually were, and it is almost laughable how overwhelmed I was at first, looking at this entirely new microcosmic

world of insects.

I had great trouble with the iButtons that were used because they were only compatible with Windows, and since I have a Mac, I struggled to find a Windows computer or laptop in order to start them recording. I had a similar issue with BORIS, as only a beta version of behavioural observation software was available for Mac, but I eventually decided against using it, since it added more time than it saved. The behaviours I was analysing were simple enough to go through directly from video and note down quickly, and in many recordings multiple nests appeared in the same frame, so I had to rely on the human eye rather than a software tool that would have required constant correction. The hotel staff were kind enough to let me use their computer, which was so old and worn out that I spent over two hours trying to get it to work, but to no avail, and they had to make other arrangements the next day by replacing the monitor and cleaning the CPU, which had accumulated eons of dust. Eventually, I was able to get the devices running and began collecting mock data as well. It is funny to realise that the thing that most defined my thesis (soil temperature) was the one I paid least attention to at the start; I initially thought the soil depth profile would simply add another interesting dimension rather than become the central theme of the first part of my thesis. But I am rushing ahead again.

In my early March days, I noticed many crickets sun-bathing on the dark turrets or tumuli excavated by the bees, and I thought that perhaps these darker structures allowed them to heat up faster than the surrounding soil. Whenever I saw bumblebees in the field, alongside the perpetual swooping swallows during their mating season, chirruping and moving in groups over the fields, I felt reassured, because it meant *H. scabiosae* would likely be active that day—since if bumblebees could heat up, so could *H. scabiosae*. I also noticed that many nests were embedded within vegetation, particularly Persian clover and buttercup patches, which seemed more protected from rain. I initially thought that later in the season they might shift toward more exposed nests in areas less influenced by vegetation in order to gain more heat, although I now realise that as summer progresses most vegetation disappears anyway and conditions become extremely hot and almost unbearable. Some iButton Hygrochron and ThermoChron data loggers did fail, but this did not deter me, because I believed I had enough alternatives and initially assumed I would not need all of them, just one or two per site. Oh, how wrong I was. On 29 March (Sunday) at 15:39, I wrote: “My dad told me to get something precautionary for bee stings, which I agree with. I do seem to run the risk of clashing with a foraging bee.” This is important because it was then that I looked up what I should do in the unfortunate event of being stung. On the first day

of my fieldwork, I was indeed stung twice—once on my arm, which was quite painful, and another time when a bee entered my shoe while I was walking through vegetation, which sounds worse than it actually was. For the first sting, I panicked, dropped the camera I was holding, and rushed to my hotel room (since it happened right outside my room rather than at the field site). I then tried desperately to remove the stinger using a credit card-like object; having not found my debit card, I grabbed something random, most likely the curtain. Seeing no visible stinger, I simply washed the area with soap and water, and I was fine. I was very relieved not to be allergic, and after that day my fear decreased significantly. After returning to Mytilene, I decided to take matters into my own hands since it was my own thesis, and began trying to acquire action cameras by borrowing from friends, Erasmus students, or locals, or even renting them, because I believed they were far superior in quality and far less cumbersome than the CCTV-style cameras whose wires could easily come loose if not handled carefully and whose focus could be affected by the slightest movement. After asking around, I obtained two cameras, which was enough for me at the time since I also had my phone and iPad, and I later decided to invest in a GoPro to get a wide-angle view. However, I never anticipated that Michael would end up lending me three of his cameras during fieldwork, two of which I used to record my chill coma recovery assays and to film myself performing them, simply as memories and for the mini-documentary I had in mind. In the end, everything worked out on that front as well, despite how much I panicked at the beginning when no one was responding to my requests and messages.

When I used to sit in Thomas' lab, which was a shared space with another professor, I would share the space with Jelle, who gave me such valuable insight about *H. scabiosae* by telling me that it exhibited a form of competitive exclusion with *H. sexcinctus*, and that both species co-occur in the western Mediterranean. He also explained that *H. scabiosae* prefers plants from the family Asteraceae, and that one of the easiest ways to distinguish it from *H. sexcinctus* is the absence of mite pockets (often referred to as acarinarium), specialised morphological structures used to house and transport symbiotic mites. This made me worry for a while whether I was actually observing *H. sexcinctus* or *H. scabiosae*, and when he later confirmed after visiting my field site in April that it was indeed *H. scabiosae*, my joy knew no bounds, as it meant everything was on the right track. I tried recording environmental variables from the Agia Paraskevi weather station every day, but during fieldwork days it was sometimes down, and in the end there was so much of my own data that adding additional environmental datasets would have only made the thesis messier, so I decided against incorporating it. I also learned how to organise all of this data in the midst of fieldwork by writing explicit step-by-step instructions for myself to access websites, often in the simplest possible form,

such as “click here,” “scroll down here,” and similar notes, because I knew I would forget once actual fieldwork began, when there was simply too much on my mind. It was good that I anticipated this, because I did end up forgetting almost everything I had prepared in March during the transition into actual fieldwork.

“The mountains are clad in shadowy muslins, seductive as the clouds part, their bare interior reflects a million different rays.” — 28 March, 1:41 pm.

15 April: 9:46 am (day 1 of sampling): “Taking my first sample soon. A bit nervous”—after Johnny helped set up my cameras and tripod and we tried out different angles to record the previous day’s setup, which really boosted my confidence. Having someone to throw ideas around with made a big difference, and he did an excellent job guiding me through different configurations. I remember us even using our shadows to check whether the tips of the shadows appeared in the video frame as a rough way of estimating the field of view. I really owe him for helping me feel comfortable on the first day of fieldwork. I started recording from 10 am because the previous day we had noticed activity around 9:40 am. However, from the seventh day onwards, when Victor asked me how I knew for certain that nothing happened around 9 am on other days, I had no answer. So I started getting up earlier and eventually realised that activity actually began in the faintest form around 8:40 am. From then on, starting recording at 9 am made more sense, effectively adding an extra hour of field time from day seven onwards. By that point, I had grown into the routine and did not mind it at all. At the end of field days I was usually completely exhausted, and after a swim in the pool, some exercise, and playing with the two sweetest newly parent dogs—Hannah and Frodo—and their puppies, who were often alone in their large enclosures shared with African emus, exotic geese, and ducks, followed by some solo football, I would quickly eat dinner, go for a walk with others to Skala Kallonis bee-neath skies where clouds formed fantastically contoured shapes in the moonlight over the stolidly still nearby mountains, and then crash on my bed without thinking about anything or looking at my phone for long. In hindsight, these were easily the best two weeks I have had on Lesvos so far, hands down.

16 April; 10:03 am (day 2 of sampling): “No activity as yet. It is a bit more chilly today compared to yesterday. Hopefully it does not rain. Even the honeybees are not out yet, as they are usually active for most of the day, staying out quite late in the evening and emerging early before everyone else in the morning. Yesterday I saw an interesting behaviour: grasshoppers displaying atop nest mounds. Swallows and mosquitoes are out. Everything is very quiet on the soil. Not a single insect in sight. Mornings like these, beautiful as they may be, replete with bird chirrups and the fresh scent of mist-

laden mountains, scare me for the beings nestled snug in the soil, as it feels cold and treacherous to venture out.”

17 April (day 3 of sampling)—I wrote the reflections for field day 1 on field day 3. I had initially planned to write reflections for each day at the end of each field day, but I was too tired and preoccupied with other tasks, such as extracting environmental data from the previous day from the Meteo Aegean online meteorological database, and taking light intensity measurements after every 10-minute recording session at both sites using a lux meter app on my phone, both of which I ultimately did not use in my analysis. It is funny to think that in the moment these tasks feel crucial, yet in the bigger picture they often do not hold up, while the smallest and seemingly trivial details, like digging soil for the thermochrons using scissors and measuring their depths with a scale, end up mattering the most. Anyways, here are my reflections for field day 1:

“Scared and truly alone, starting out proved a lot gutsier than I thought. I started doubting myself and called Johnny to confirm the previous day’s onset activity. Reassured, I started the recordings with a ball in my stomach at 10 am. Looking back, 14 April was a marvellous day. The first camera angles we got were exceptionally good (for field site 2), and now, on my third day, time, overuse, and shifting conditions had already begun to strip every tripod setup of its original charm. I still think those first-day angles were somehow different: everything shimmered less, nest entrances were clear as daylight, and clarity reigned supreme. After two recordings, shocked and aghast, I saw Sveta’s camera die during the third session, and ever since (it has only been a day), I change it every second recording cycle because I assume it only lasts for two 10-minute sessions and I would not dare experiment with whether it survives a third, especially at this stage of fieldwork. I did test it earlier in the island’s lounge setting, where it lasted longer, but I had failed to factor in the intense heat of the sun beating down on the equipment. All in all, after two bee stings and two mistakes (one being forgetting to switch on, or rather check, Thanasis’ camera), it was disastrous, but still a learning curve.”

On my day 4 of sampling (18 April), Victor found *H. scabiosae* and *L. malachurum* on the hotel premises, right outside my room and the lab, which made it convenient to collect individuals and carry out trials. I was relieved to be able to work with these individuals there rather than disturb my behavioural sites for the chill coma recovery assay experiments, which Michael later helped me set up on the spot after I felt somewhat dejected when the critical thermal limits machine broke down. This idea came together quickly, and when he suggested it, I was glad it aligned so well with the behavioural component of my work, so that the two parts began to feel like

chapters of the same story told from different perspectives, ultimately integrating into a single framework, which felt quite exciting. And there were so many individuals of *L. malachurum* there that I did not need to worry about double-counting. I was more concerned about *H. scabiosae*, but then I remembered Victor's summer sites, and, as it turned out, new *H. scabiosae* nests had begun appearing there as well. This meant I could also procure additional individuals from those locations, as well as from the small site between my Field Sites 1 and 2 in times of desperate need. I began writing less and less in my field journal as the days went on, partly because I often left it on the ground near the sites, where it camouflaged so well with the soil that I sometimes forgot it was even there. My field sessions consisted of recording for the first 10 minutes of every hour from 9 to 6 pm, followed by 5-minute intervals in which I measured light intensity using a lux meter app on my phone, and then quickly running to collect bees for my chill coma recovery assays. Each trial took variable time, so achieving a sample size of 20 individuals per species required extreme speed and efficiency in my setup. For this reason I relied on voice recordings rather than written notes, narrating what was happening, how long each bee had been in chill coma, and whether chill coma had been induced, while simultaneously preparing equipment such as the thermal camera, thermocouples, GoPro, tweezers positioned near the measuring scale, and blank sheets for photographing and recording measurements. At the start, I recorded the bee weight (with vial), and I would then hold the thermal camera above the bee in chill coma and continue narrating what I observed, including duration, temperature, and recovery milestones such as righting and take-off, before finally recording the vial weight. It was extremely hectic, and I sometimes reduced lunch to 10–15 minutes because the activity window was short and I wanted to maximise sampling before 3 pm, after which chill coma recovery assays were no longer feasible due to the lack of active bees. In addition, the iButtons were not fully reliable due to ageing batteries, so I had to excavate them from the soil (5–10 cm deep depending on placement), extract the data, and reinstall them at their respective depths and positions, carefully recompact the soil to avoid disturbance. Initially, I was sceptical, as I thought all temperatures would be highly correlated across depths and that a single Thermochron per site would suffice. I am now glad I was proven wrong, as this allowed me to capture much more reliable temperature variation. On alternate days, this meant field sessions often lasted until 6:30 pm. Overall, this taught me how to commit fully to field time and accept delayed rewards, but I genuinely enjoyed it, especially being surrounded by equally passionate people like Michael, Victor, and Isabella, who helped me keep going. My final field day notes were quite sad; I was genuinely reluctant to leave, and I could not believe that this intense period would soon fade. I knew that returning to Mytilene would make

life feel dull again, and it did—but only for a while.

24 April; Friday (day 10 of sampling): “This has become my life now; alas, it all crescendos to an end. As all good things eventually do, it is coming to an end, and it will be really hard adjusting to life back in Mytilene. I have come to realise how much I like this phase of life: the rush that is persistently swelling, and how involved I am in things other than myself. I have come to see a family in Victor, Michael, and Isabella, and I will dearly miss them, most of all Michael. He made me feel heard after so long and tremendously helped me with equipment. I cannot believe today is the last day I spend here. Yesterday was especially miserable, full of rain and no activity, but I am still glad I could catch three individuals for my take-off study. I now have 10 good samples for each species, and I plan to do five more ideally for each species, though the best would be 20 individuals. My plan is to sample the other sites as best as I can today and to include individuals from my own site for the final day. Yesterday I had a funny moment when I was essentially waiting for a large cloud to pass, because bees come out of their nests on cold days only when there is a surge of sunlight breaking through, creating a brief glorious window that cannot be missed. It made me realise how much I take things for granted, and how in nature even the finest behaviours are governed by rhythms that stop for no one and proceed at their own pace, entirely bee-yond my control. I was at an *H. scabiosae* nest where Victor had sampled in previous years, and I saw an individual peep out of its nest and remain there.”

And thus, as unceremoniously as it had begun, my field journal ends here. Having learnt so much through this experience, the three pages I had initially dedicated to reflections have grown to twenty-two, and I still feel I have more to say. When I first began this master's, I came from a strongly quantitative background, and I still do, but I have learnt so much from my peers over the past two years in ways that have truly changed how I think about and practise science. Coming from a background where I had to secure scholarships to study and explore outside my country, I was deeply motivated to gain a place in a programme like this. I was initially rejected for the scholarship and placed on the waiting list, but eventually, after a withdrawal, I was offered the place and was able to join. The experience exceeded anything I had ever imagined. It helped me slow down just enough, look beyond my immediate field, and learn from other disciplines while still engaging deeply with my own. I have learnt to recognise my own prejudices and biases, and to accept that any study, however objective it may appear, carries a deeply subjective motivation that shapes it in subtle ways. There was also a considerable amount of subjectivity in this process, for example in estimating bee counts—whether there were 10–15 or 15–20 individuals—since I

simply did not have the time to count every single one. I had to rely on intuition in ways I never expected in a scientific project, but here I was, estimating abundances across time points, sometimes easily and sometimes with difficulty, yet consistently using the same margin of error; in this way, the broad behavioural component of the study gradually took shape. I panicked for the first time when I began data analysis, because after accumulating over 500 GB of data (and navigating several external hard drive failures), I finally had everything in place and analysis was all that remained. Four full days in the island's lounge, and working until 2 a.m. when I returned to my room each night, were required to push through this phase and move on to the results section. It was by far one of the hardest parts of the project, because although I could endure long hours in the field, staring at a screen for over 12 hours a day was something my eyes were neither prepared nor equipped for. Nevertheless, believing that one must sweat more in practice so that one bleeds less in battle, and with sticky notes lining my deadlines and people holding me accountable, I managed to complete it by a narrow margin and still find time to write this long reflection section, which I felt was necessary to do justice to what this programme has given me. I felt a sense of accomplishment unlike anything before, and the backdrop of bee-leaguered rejections from internships, research assistantships, and PhD applications did not deter me during this phase. This thesis taught me to think and act independently, with my supervisors serving as crucial fulcrums who guided my direction while leaving the details to me, for which I am deeply grateful, as I have come to understand the extent of my own independence. My limits were truly tested throughout this thesis, and I am glad they were, as I learnt a great deal from the experience. Although I had already encountered extensive ecology and evolution theory through my courses at RUG, each covering fascinating and diverse topics, it was only when I was in the field actively trying to understand why something was happening that I felt another part of my thinking truly activate for the first time, one that began identifying gaps in the project and considering how to improve or address them. This is why the future directions section of the thesis became so extensive, as many of these ideas emerged organically over time through doing the work itself, rather than from planning in advance.

Throughout these two years, I have been trained to think and act analytically, and I have learnt so much about the extraordinary world of ecology and evolutionary biology that I am certain a future in this field would make me happy. I have been shaped by so many people from different walks of life, and I carry a part of them wherever I go. The way I speak now, the way I choose my words more carefully, the way I listen, the way I carry myself, the way I get up after countless rejections, and the way things affect me have all changed. During my MITACS internship in Canada, Jacob, one of the master's students

there, told me to “pick the things worth getting upset about,” and I feel I have fully taken that on board through this master’s. I care more deeply about the things I truly care about, and I have stopped worrying about things I cannot control. I have learnt to navigate life—not just academically, but socially as well: adapting to places, to people, to the way they think, what I can learn and improve, and what I can safely ignore. I feel this programme has been a kind of training ground for life before stepping into a world of intense competition, where personality often seems to matter less than what is written on paper, whereas this programme has taught me to value the opposite. I am deeply grateful for the opportunity and the privilege of sharing this journey with an extraordinarily inspiring and warm cohort and endearing juniors, who made my time in Lesvos truly memorable, along with everything that Groningen, Gran Canaria, and Lesvos have taught me. That bee-ing said, I still have my last months in Lesvos ahead, and I will not let sentimentality take over and prevent me from enjoying them fully. I hereby write the final words I would write for my Islands and Sustainability master’s journey as a student. As an alumnus, I am sure I will keep coming back to this small island of ours, because that is what this programme and its experiences do: they brand you as an islander for life.